

# Activity 1: Build an EMI Calculator using AI

## Purpose

Learn how to use AI as a coding assistant to generate a simple Oracle JET component. The generated EMI Calculator will be used throughout the remaining Day 2 activities to understand, debug, enhance, document, and review AI-generated code.

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## Pre-requisites

- VS Code installed
  - Cline extension configured
  - AI provider configured in Cline
  - Oracle JET application created
  - Basic knowledge of Oracle JET and KnockoutJS
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## Estimated Duration

45–60 Minutes

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## Scenario

You are part of the SmartBank development team. The business team has requested a simple **EMI Calculator** that allows customers to estimate their monthly loan installment before applying for a loan.

Use AI to generate the component and review the generated code before proceeding.

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## Tasks

### Task 1: Generate the EMI Calculator

Use AI to generate the complete frontend component.

### **Suggested Prompt**

Create a simple EMI Calculator component using Oracle JET and KnockoutJS.

Requirements:

- Use Oracle JET components and Redwood theme.
- Generate only the frontend.
- Do not use any backend or REST APIs.
- Do not use external libraries.

The page should contain:

- Loan Amount
- Annual Interest Rate (%)
- Loan Tenure (Years)
- Calculate EMI button
- Monthly EMI result

When the user clicks Calculate EMI:

- Validate that all values are greater than zero.
- Calculate the Monthly EMI using JavaScript.
- Display the EMI rounded to two decimal places.

Use KnockoutJS observables and functions.

Keep the code clean, simple, and beginner friendly.

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## **Task 2: Run the Application**

Run the Oracle JET application and verify that:

- All input fields are displayed.
  - The Calculate EMI button is visible.
  - EMI is calculated correctly.
  - Invalid inputs are handled appropriately.
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## **Task 3: Review the Generated Code**

Use AI to review the generated code.

### **Suggested Prompt**

Review the generated EMI Calculator.

Check for:

- Code readability

- Oracle JET best practices
- KnockoutJS best practices
- Possible improvements

Do not modify the functionality.

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## Expected Deliverables

By the end of this activity, you should have:

- A working EMI Calculator
- Oracle JET UI
- KnockoutJS ViewModel
- EMI calculation logic
- AI-generated code review recommendations